1 My solutions for exercises

1.1 Exercise 3.7

Prove the Axiom of Pair, the Axiom of Union, and the Axiom of Power Set using their weaker versions.

1.1.1 The Axiom of Pair

Weak Axiom of Pair. For any A and B, there is a set C such that $A \in C$ and $B \in C$.

Proposition. For any A and B, there is a set C such that $x \in C$ if and only if $x = A$ or $x = B$.

Proof. Let A and B be sets, and let C be a set such that $A \in C$ and $B \in C$. Now consider the set $Y = \{x \in C : x = A \text{ or } x = B\}$. We show that $x \in Y$ if and only if $x = A$ or $x = B$. Indeed, $x = A$ or $x = B$ for any $x \in Y$. Conversely, if $x = A$ or $x = B$, then $x \in C$ by the definition of C, giving $x \in Y$. Hence Y has the required property.

1.1.2 The Axiom of Union

Weak Axiom of Union. For any S, there exists U such that if $x \in A$ and $A \in S$, then $x \in U$.

Proposition. For any S, there exists a U such that $x \in U$ if and only if $x \in A$ and $x \in S$ for some A.

Proof.

Let S be a set. By the weak axiom of union, there exists a U such that for every x and A, $x \in U$ if $x \in A$ and $A \in S$. Define $Y = \{x \in U : x \in A \text{ and } A \in S \text{ for some } A\}$. Clearly, if $x \in Y$, then $x \in A$ and $A \in S$ for some A. Conversely, if x is a set such that $x \in A$ and $A \in S$ for some A, then $x \in U$, by definition. Hence $x \in Y$.

Therefore, For any S, there is a Y such that $x \in Y$ if and only if $x \in A$ and $x \in S$ for some A, which is the desired conclusion.

1.1.3 The Axiom of Power Set

Weak Axiom of Power Set. For any set S, there exists a P such that $x \subseteq S$ implies $x \in P$.

Proposition. For any set S, there exists a P such that $x \in P$ if and only if $x \subseteq S$.

Proof. Let S be a set, and let P be the set implied by the weak axiom of power set. Consider the set $Y = \{x \in P : x \subseteq S\}$. Since $x \subseteq S$ implies $x \in P$, the statements "$x \subseteq S$" and "$x \in P$ and $x \subseteq S$" are equivalent. Hence we have $x \in Y \leftrightarrow x \in P$ and $x \subseteq S \leftrightarrow x \subseteq S$. Therefore Y satisfies the required property.

1.2 Exercise 4.4

Let A be a set; show that a "complement" of A does not exist. (The "complement" of A is the set of all $x \notin A$.)

Proof. Let A be a set. Suppose that there exists a set X such that $x \in X$ if and only if $x \notin A$. Now consider the set $A \cup X$. For every x, $x \in A \cup X$ if and only if $x \in A$ or $x \in X$, i.e., if and only if $x \in A$ or $x \notin A$. That implies $x \in A \cup X$ for every x and thus $A \cup X$ is the universal set. But this contradicts Exercise 3.3, which states no universal set exists. Therefore the "complement" of A does not exist.

1.3 Exercise 4.5

Let $S \neq \varnothing$ and A be sets.

1.3.1 (a)

Set $T_1 = \{Y \in \mathcal{P}(A) : Y = A \cap X \text{ for some } X \in S\}$, and prove $A \cap \bigcup S = \bigcup T_1$ (generalized distributed law).

Proof. First, we introduce a small lemma:

Lemma. For every $X \in S$, $A \cap X \in \mathcal{P}(A)$. Indeed, $A \cap X \subseteq A$, and so by definition of $\mathcal{P}(A)$, $A \cap X \in \mathcal{P}(A)$.

Now we prove that both sets are equal using the Axiom of Extensionality.

First, let $x \in A \cap \bigcup S$. To show that $x \in \bigcup T_1$, we seek an element of T_1 which contains x. since x belongs to $\bigcup S$, $x \in X$ for some $X \in S$. Now consider the set $A \cap X$. By the above lemma, it is an element of $\mathcal{P}(A)$, and thus an element of T_1. Since A and X both contain x, $x \in A \cap X$. Therefore, $A \cap X$ is the set we were seeking.

Conversely, let $x \in \bigcup T_1$. By the Axiom of Union, there exists a $Y \in T_1$ which contains x. Now by the definition of T_1, $Y = A \cap X$ for some $X \in S$. Since $x \in Y = A \cap X$, we have $x \in \bigcup S$, as $x \in X$ for some $X \in S$. Hence $x \in A \cap \bigcup S$, as desired.

Therefore, the Axiom schema of Extensionality, $A \cap \bigcup S = \bigcup T_1$.

1.3.2 (b)

Set $T_2 = \{Y \in \mathcal{P}(A) : Y = A - X \text{ for some } X \in S\}$, and prove
$A - \bigcup S = \bigcap T_2$
$A - \bigcap S = \bigcup T_2$.
(generalized De Morgan laws).

1.4 Exercise 4.6

Prove that $\bigcap S$ exists for all $S \neq \varnothing$.

Proof. Let S be a non-empty set. We show that $Y = \{x \in \bigcup S : \forall A(A \in S \to x \in A)\}$ is the desired set. First, let $x \in Y$. Then by the definition of Y, $x \in A$ for every $A \in S$. Conversely, let x be a set such that for all $A \in S$, $x \in A$. Since S is non-empty, there exists an A such that $A \in S$. Then by the hypothesis, $x \in A$, and so $x \in \bigcup S$. Therefore $x \in Y$.